

Estela Pérez Luque, PhD

Human Factors Engineer

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PROFILE

Human Factors Engineer with 6+ years of experience in ergonomics, digital human modelling, simulation, and validation studies in complex engineering environments. My background combines user-centred design, human-system interaction analysis, and evaluation of posture, motion, usability, and anthropometric variability to support better design decisions. I have worked closely with Volvo Cars, Volvo Trucks, and Scania, translating user feedback, simulation results, and technical constraints into requirements, validation activities, and design recommendations. I am particularly motivated by workstation and interface design challenges where physical and cognitive ergonomics directly shape safer and more effective operator performance.

TECHNICAL SKILLS

Core tools: MATLAB, Python, Excel

CAD & Simulation Tools: IPS IMMA, Siemens Jack, AnyBody, Autodesk Inventor, SolidWorks

Methods: V&V, simulation-based evaluation, model benchmarking, experimental design, motion capture, time-series analysis, requirements support

Human Factors: physical and cognitive ergonomics, HMI, usability, task analysis, anthropometric variability, visibility, reachability, posture and seating analysis

Languages: English (C1), Spanish (Native), Swedish (B1)

PROFESSIONAL EXPERIENCE

Human Factors Researcher - University of Skövde, Sweden (Onsite)

Sep 2020 – Oct 2025

- **End-to-end Human Factors evaluation:** Designed and executed Human Factors studies from early operational and ergonomic analysis to validation across multiple vehicle configurations, user populations, and operating conditions, providing structured evidence for design decisions.
- **Anthropometric modelling and sampling:** Built representative user populations using percentile-based selection, anthropometric ellipsoids, and boundary-case manikins, improving accommodation assessments across 199+ participants and multiple datasets.
- **Digital human modelling and simulation:** Used IPS IMMA, Siemens Jack, and AnyBody to assess posture, reachability, and physical ergonomics; ran multi-objective optimisation across 1,000+ configurations and 14 digital human models, replacing manual trial-and-error with scalable, data-driven evaluation.
- **Usability and validation:** Evaluated usability and operator interaction through structured analysis, user studies, and iterative validation of predicted versus observed behaviour, reducing subjective assessment and supporting more robust design reviews.
- **Stakeholder collaboration and traceability:** Structured technical constraints, evaluation criteria, and Human Factors inputs with Volvo Cars, Volvo Trucks, and Scania, improving traceability and alignment across multidisciplinary teams.
- **Research and knowledge transfer:** Delivered outcomes through 20+ publications and 15+ international presentations, helping increase adoption of simulation-supported, data-driven Human Factors methods in research and industry.
- **Teaching and technical instruction:** Taught CAD and digital human modelling (DHM) tools in ergonomics to Bachelor's and Master's students, strengthening practical understanding of simulation-based Human Factors assessment and user-centred design methods.

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Visiting Researcher - University of Antwerp, Belgium (Onsite)

Sep 2024 – Mar 2025

- **Experimental campaign leadership:** Led full-cycle data collection for a large-scale, multi-year study, including ethics approval, protocol design, recruitment, and multi-sensor acquisition, enabling structured validation across datasets and institutions.
- **Validation data quality and reproducibility:** Designed motion capture and 4D body scanning protocols for 30+ participants and built reproducible pipelines, improving data quality, repeatability, and cross-study comparability.

Co-founder and R&D Lead - Swe-Ex AB, Sweden (Startup)

Sep 2021 – Mar 2023

- **Entrepreneurship:** Co-founded a startup developing wearable safety support equipment for industrial environments, bringing human factors into early product development.
- **Human Factors and requirements definition:** Translated user needs and safety requirements into measurable Human Factors requirements and evaluation criteria, guiding three prototype iterations from CAD concept to physical and ergonomic validation.
- **Prototype validation and iterative design:** Combined user testing with technical assessment to evaluate fit, usability, and safety, enabling faster iteration and more evidence-based design decisions.

Research Assistant - University of Skövde, Sweden (Onsite)

Jun 2019 – Sep 2020

- **Applied Human Factors with industry partners:** Developed simulation studies and ergonomic analyses across 6+ industry projects, producing reports, demonstrations, and recommendations that supported early decisions.
- **DHM knowledge transfer:** Created tutorials, educational materials, and training content on digital human modelling tools, improving adoption and more effective use of simulation-based ergonomics methods.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis:](#) Human Posture and Motion Prediction for Automotive Ergonomics Design

Focus: digital human modelling, simulation-based evaluation, V&V, anthropometric variability, ergonomics for complex product development

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

LEADERSHIP & RECOGNITION

- Elected Student Representative and Board Member in doctoral committees, representing 40+ PhD students in university governance and program improvement decisions.
- Recipient of multiple international grants and awards, including travel and conference funding supporting research dissemination and cross-institutional collaboration.

RESEARCH PUBLICATIONS

Full publication list available at: [Google Scholar – Estela Pérez Luque](#)